

Code No: R22A0507

MALLA REDDY COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous Institution – UGC, Govt. of India)

II B.Tech II Semester Supplementary Examinations, November 2024**Object Oriented Programming through Java**

(CSE,IT,CS&IT,CSE-CS,CSE-AIML,CSE-DS,CSE-IOT & B.Tech-AIML)

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Time: 3 hours**Max. Marks: 60****Note:** This question paper contains two parts A and B

Part A is compulsory which carries 10 marks and Answer all questions.

Part B Consists of 5 SECTIONS (One SECTION for each UNIT). Answer **FIVE** Questions, Choosing ONE Question from each SECTION and each Question carries 10 marks.

<u>PART-A (10 Marks)</u>			BCLL	CO(s)	Marks
<u>(Write all answers of this part at one place)</u>					
1	A	What is the property used to know the length of an array?	L1	CO-I	[1M]
	B	Define this pointer in Java.	L1	CO-I	[1M]
	C	List different types of inheritance.	L1	CO-II	[1M]
	D	When do we use the implements keyword?	L1	CO-II	[1M]
	E	What is the benefit of handling exceptions?	L2	CO-III	[1M]
	F	What is inter-thread communication?	L1	CO-III	[1M]
	G	What is a file?	L1	CO-IV	[1M]
	H	Define collection class?	L1	CO-IV	[1M]
	I	Give examples of an event.	L1	CO-V	[1M]
	J	Write syntax to create a JLabel swing component.	L1	CO-V	[1M]
<u>PART-B (50 Marks)</u>					
<u>SECTION-I</u>					
2		Discuss operators in Java with suitable examples.	L2	CO-I	[10M]
		OR			
3	A	Compare and contrast class and object with a suitable program.	L2	CO-I	[5M]
	B	Develop a Java program to find the factorial of a given number using recursion.	L3	CO-I	[5M]
<u>SECTION-II</u>					
4	A	Explain the usage of the final keyword with suitable examples.	L2	CO-II	[5M]
	B	Discuss abstract class in Java in detail.	L2	CO-II	[5M]
		OR			
5	A	Discuss interfaces in detail with suitable examples.	L2	CO-II	[5M]
	B	Create a package named 'company' that contains the Employee class representing employee details. Include methods for setting and retrieving employee information such as name, ID, and salary. Then, showcase the	L3	CO-II	[5M]

process of importing this package into another Java class and demonstrate the utilization of the Employee functionalities.

SECTION-III

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| 6 | Discuss the different keywords of exception handling in Java in detail with suitable examples. | L2 | CO-III | [10M] |
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OR

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| 7 | A Explain the life cycle of a thread with a neat diagram. | L2 | CO-III | [5M] |
| | B Write a Java program to demonstrate thread priorities. | L3 | CO-III | [5M] |

SECTION-IV

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| 8 | A Write a Java program to demonstrate ArrayList. | L3 | CO-IV | [5M] |
| | B Discuss file management using File class. | L2 | CO-IV | [5M] |

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| 9 | Discuss the different types of JDBC drivers, including the advantages and disadvantages of each type, accompanied by clear diagrams. | L2 | CO-IV | [10M] |
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SECTION-V

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| 10 | A Briefly discuss AWT class hierarchy with a neat diagram. | L2 | CO-V | [5M] |
| | B Explain the need for a layout manager in Java. Write a program to demonstrate the use of a layout manager. | L3 | CO-V | [5M] |

OR

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| 11 | A Discuss Adapter classes and it's role in detail. | L2 | CO-V | [4M] |
| | B Develop a Java swing application to handle mouse events. | L3 | CO-V | [6M] |
